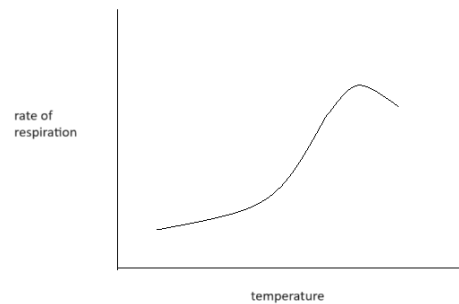


1(a)(i)

- 1 correct axis labels ;
 - 2 line rises and then falls ;
 - 3 line rising at an increasing rate to the optimum ;
- e.g. three marks for:



- 3 e.g. x-axis = temperature
y-axis = rate of respiration

ignore anaerobic / aerobic

3 **ignore** absence of fall
accept any curve with an increasing gradient

mp2 and mp3 **ignore** starting point of curve

ecf for incorrect mp1 with correct axes labels



Question	Answer	Marks	Guidance
1(a)(ii)	any eight from: any eight from: 1 use a minimum of five stated temperatures ; 2 suitable <u>method</u> to maintain a constant temperature ; 3 use, same / stated, number / amount / mass, of seeds ; 4 use, same / stated, species / type / variety, of seeds ; 5 use, same / stated, volume / concentration, of sodium hydroxide / NaOH ; 6 <u>method</u> to, equilibrate / acclimatise, <u>seeds</u> to new temperature ; 7 at each temperature, measure the distance (moved by the coloured liquid), in a, set / stated, time or at each temperature, measure the time, (for the coloured liquid), to travel a, set / stated, distance ;	8	1 temperatures must include 10 °C and 50 °C must see units at least once ignore temperatures outside the range 2 A water-bath / incubator ignore temperature-controlled room / air conditioning / heater / fridge A <i>ref.</i> to use thermometer and ice / warm water 4 Ignore age (of seeds) 5 Ignore mass / amount mp3, mp4 and mp5 allow standardised / fixed / controlled / constant as AW for same ignore known 6 A leave seeds in respirometer at new temperature / in water-bath / AW, for, set / stated time if stated, minimum time 5 minutes 7 allow note / record as AW for measure ignore calculate as AW for measure allow length / mm as AW for distance ignore <i>ref.</i> to measuring without sodium hydroxide

Question	Answer	Marks	Guidance									
1(a)(ii)	8 (replace seeds / repeat, with) sterile/ inert / glass / plastic / AW, beads or dead / AW, seeds ; 9 at each temperature, take at least three measurements ; 10 safety comment with hazard and risk and precaution ; <table><tr><th>hazard</th><th>risk</th><th>precaution</th></tr><tr><td>NaOH / sodium hydroxide</td><td>irritant / corrosive / caustic / damage to eyes</td><td>(wear) gloves / safety glasses / goggles / PPE</td></tr><tr><td>seeds</td><td>irritant / allergy</td><td>(wear) gloves / PPE</td></tr></table> 11 <i>Idea of:</i> open the tap and use the syringe to reset the coloured liquid ;	hazard	risk	precaution	NaOH / sodium hydroxide	irritant / corrosive / caustic / damage to eyes	(wear) gloves / safety glasses / goggles / PPE	seeds	irritant / allergy	(wear) gloves / PPE		allow how far as AW for distance allow how long as AW for time ignore collecting gas in the syringe ignore <i>ref. to</i> open / close, tap 8 ignore respirometer with no seeds R if another variable is changed as well as the seeds 9 ignore calculate a mean A three replicates / repeat 2 times / multiple times, at each temperature 10 ignore burns ignore biohazard ignore hot water as a risk
hazard	risk	precaution										
NaOH / sodium hydroxide	irritant / corrosive / caustic / damage to eyes	(wear) gloves / safety glasses / goggles / PPE										
seeds	irritant / allergy	(wear) gloves / PPE										

Question	Answer	Marks	Guidance
1(a)(iii)	any four from: 1 divide diameter by 2 (to give radius) or radius = 0.5 mm ; 2 calculate (cross sectional) area (of the capillary tubing) or area = 0.79 mm² ; 3 distance multiplied by (cross sectional) area (to calculate volume of gas) ; 4 divide by time ; 5 calculate a mean distance / rate ; 6 divide by mass of seeds used ;	4	2 A πr^2 or $\pi d^2 / 4$ Use calculation guidance $\pi r^2 \times$ distance moved gets mp2 and mp3
1(b)(i)	0.846 ;	1	use calculation guidance A 0.85
1(b)(ii)	(species =) B ; (evidence =) carbon dioxide (volume) is greater than oxygen (volume) ; ora	2	No ecf from 1(b)(i) A RQ is greater than 1 No ecf for wrong species
1(b)(iii)	any one from: <i>reason why oxygen measurements are too low</i> 1 (water bath) temperature was too low ; 2 duration of the test, was too short / less than one minute ; 3 <i>idea of</i> gas escaping from, respirometer / apparatus / AW ;	1	ecf for species C only in 1(b)(ii) ignore human error unqualified A temperature was high and enzymes denatured R gas coming into apparatus